



University of Asia Pacific

Department of Computer Science and Engineering

CSE 450: VLSI Lab

LAB REPORT

Experiment Number:01

Experiment Title: Design and Simulation of a 2-Input CMOS NOR Gate using Cadence Virtuoso.

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1. Experiment Name

Design and Simulation of a 2-Input CMOS NOR Gate using Cadence Virtuoso.

2. Objective

The objective of this experiment is to design a transistor-level CMOS 2-input NOR gate using Cadence Virtuoso. The experiment involves creating the circuit schematic, developing a test bench to evaluate its performance, and verifying the DC transfer characteristics and transient response of the designed circuit.

3. Apparatus / Hardware & Software Requirements

- Cadence Virtuoso Studio (IC25.1)
- PuTTY for remote Linux server access via SSH with X11 forwarding
- XLaunch as the local X server
- WinSCP for secure file transfer

4. Circuit Diagram / Schematic

The NOR gate is designed using standard CMOS logic, consisting of two PMOS transistors connected in series to form the pull-up network and two NMOS transistors connected in parallel to form the pull-down network. The inputs, V_a and V_b , are applied to the corresponding PMOS and NMOS transistor gates, while the output, V_{out} , is obtained from the common drain connection of the transistor networks.

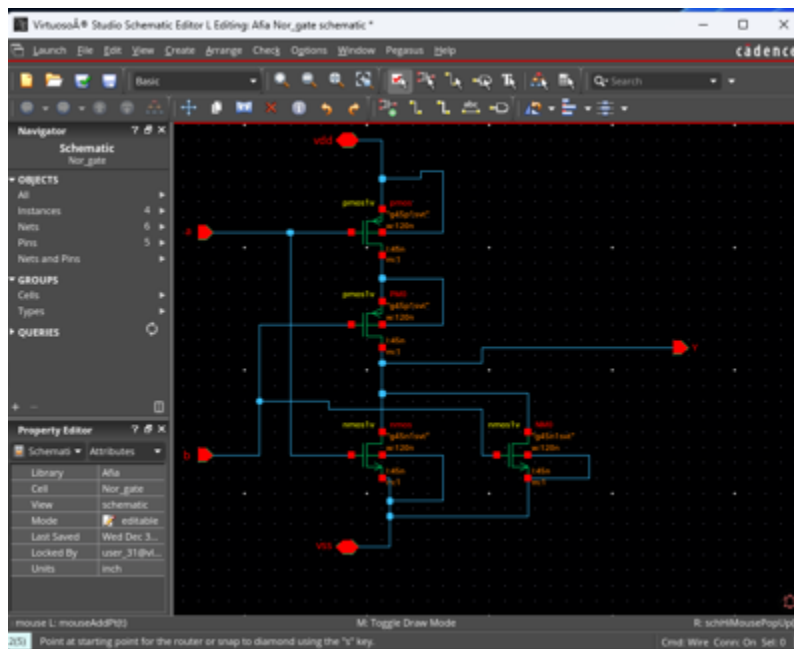


Figure 1: Transistor-level schematic of the 2-input CMOS NOR gate

A schematic symbol was created for the NOR gate, allowing it to be instantiated as a black-box component in higher-level testbenches and circuit designs.

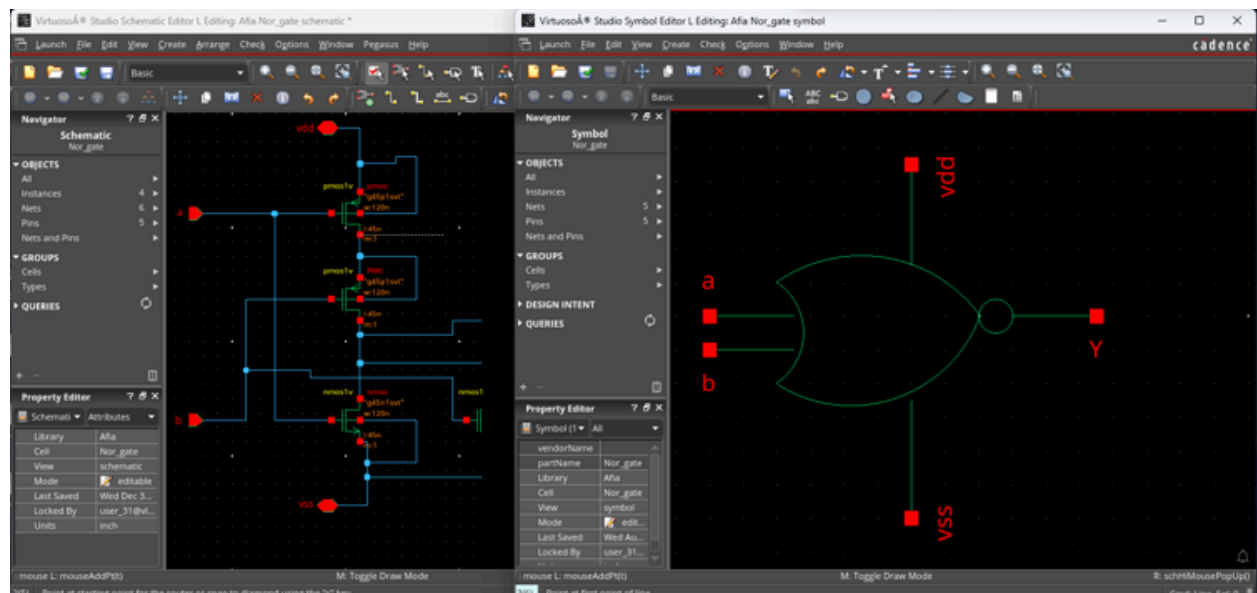


Figure 2: Schematic symbol of the designed NOR gate.

Functional verification was carried out by creating a separate testbench schematic. Two pulse voltage sources were connected to the inputs V_a and V_b , while a 1 V DC voltage source was connected to VDD. The NOR gate symbol was then instantiated in the testbench, with its output connected to Vout for observation and analysis.

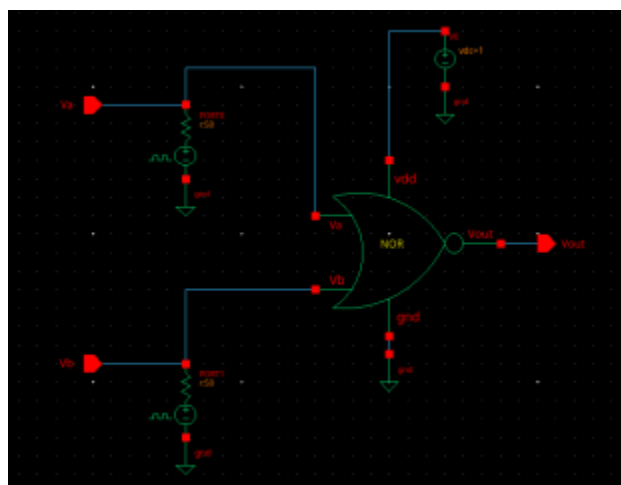


Figure 3: Testbench schematic used for DC and transient simulation.

6. Output Analysis / Observations

A 2-input NOR gate performs an OR operation followed by inversion. The output, Y, is HIGH (1) only when both inputs A and B are LOW (0). When either input is HIGH (1), the corresponding NMOS transistor turns ON, providing a low-resistance path to ground and pulling the output Y to approximately 0 V. When both inputs are LOW, both PMOS transistors turn ON while the NMOS transistors remain OFF, allowing the output to be pulled up to VDD.

Theoretical Truth Table for a 2-Input NOR Gate

Input A	Input B	PMOS Status (PM0, PM1)	NMOS Status (NM0, NM1)	Expected Output (Y)
0 (GND)	0 (GND)	PM0: ON, PM1: ON	NM0: OFF, NM1: OFF	1 (VDD)
0 (GND)	1 (VDD)	PM0: ON, PM1: OFF	NM0: OFF, NM1: ON	0 (GND)
1 (VDD)	0 (GND)	PM0: OFF, PM1: ON	NM0: ON, NM1: OFF	0 (GND)
1 (VDD)	1 (VDD)	PM0: OFF, PM1: OFF	NM0: ON, NM1: ON	0 (GND)

During the execution of this lab experiment, the simulation could not be completed successfully due to persistent environmental and technical issues within the Cadence Virtuoso environment. The following issues were encountered:

- Read-Only Lock and File Permission Errors: As shown in Figure 1, the schematic was opened in read-only mode and was locked by another user (user_31@vlsi...). This prevented necessary modifications, parameter editing, and successful netlist generation.
- ADE Simulator Engine Connection Failure: When launching the Analog Design Environment (ADE), license checkout and environment path errors occurred. As a result, the Spectre simulator could not be successfully launched, preventing the DC and transient simulations from being performed.

7. Result

The schematic design and testbench setup of the 2-input CMOS NOR gate were successfully completed using the Cadence Virtuoso Schematic Editor XL. All required components, including PMOS and NMOS transistors, power rails (VDD and GND), pulse voltage sources, and output pins, were correctly connected in accordance with standard CMOS logic design principles. However, due to software configuration, licensing, and file-locking issues in the workstation

environment, the transient simulation could not be executed, and simulated waveform plots could not be obtained. Therefore, the schematic design objective was successfully achieved, while the simulation objective remained partially unfulfilled because of environment-level limitations of the Cadence Virtuoso tool.

8. Github Repository Commits

All schematic files, testbench captures, and lab report documentation have been committed to the personal Git repository.

GitHub Repository Commits URL: [github](#)

9. Conclusion

From this experiment, I gained valuable practical experience in designing static CMOS logic circuits using complementary PMOS pull-up networks in series and NMOS pull-down networks in parallel. I also developed hands-on skills in using Cadence Virtuoso Studio Schematic Editor XL to place 45 nm transistors, configure their dimensions and manage circuit nets and pins. Furthermore, I learned how to create hierarchical symbol views such as Nor_gate2 and integrate them into a dedicated testbench schematic. The experiment also provided practical knowledge of configuring VDC and VPULSE voltage sources and establishing the required power and ground connections for transient analysis. Finally, I gained experience in troubleshooting VLSI CAD environment issues, including library path configuration, file locking, license availability, and simulator initialization.